

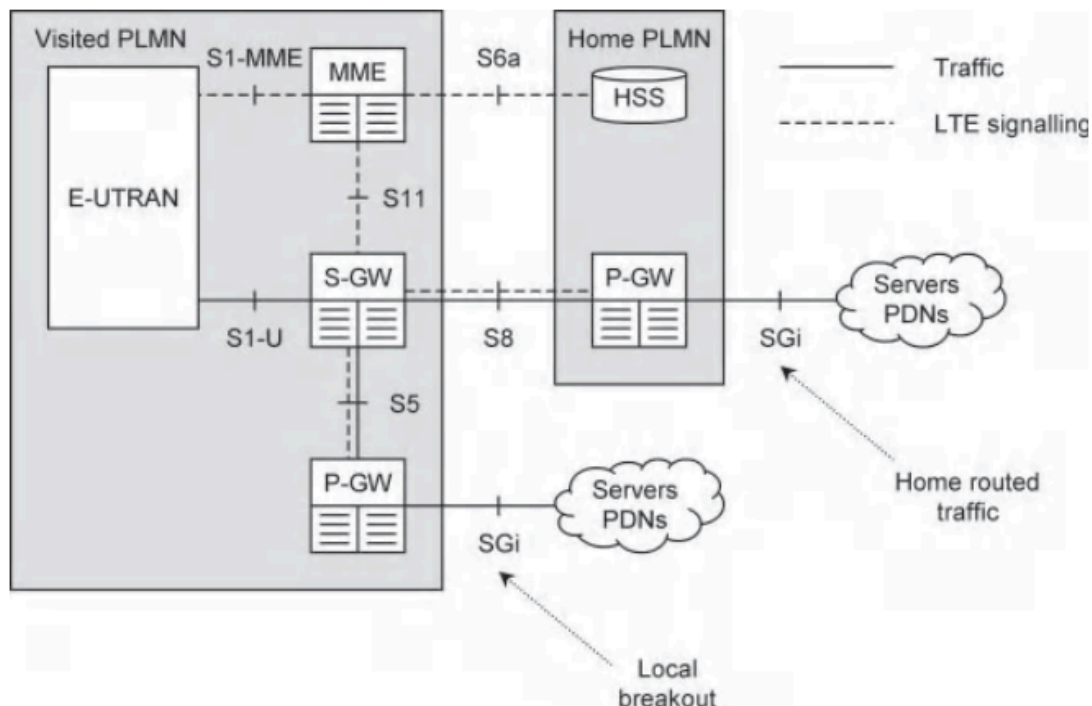


Figure 2.5 Architecture of LTE for a roaming mobile.

Here's a **clear exam-ready answer with diagram + explanation** of LTE architecture components:

LTE Architecture Components

◆ 1. E-UTRAN (Evolved Universal Terrestrial Radio Access Network)

Component: eNodeB (eNB)

- Main base station in LTE
- Connects **user equipment (UE)** to the network
- Handles:
 - Radio communication
 - Scheduling
 - Handover

◆ 2. EPC (Evolved Packet Core)

EPC is the **core network** responsible for data and control.

◆ (a) MME (Mobility Management Entity)

- Controls **signalling and mobility**
 - Functions:
 - Authentication of users
 - Tracking user location
 - Session management
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◆ (b) S-GW (Serving Gateway)

- Acts as a **data router**
 - Functions:
 - Forwards user data packets
 - Maintains data during handover
 - Connects eNodeB and P-GW
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◆ (c) P-GW (Packet Data Network Gateway)

- Connects LTE network to **external networks (Internet)**
 - Functions:
 - Allocates IP address
 - Controls data usage and QoS
 - Acts as exit/entry point
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◆ (d) HSS (Home Subscriber Server)

- Central **database of users**
- Stores:
 - Subscriber details
 - Authentication information

- Service profiles
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◆ 3. Interfaces in LTE

Interface	Connects	Function
S1-MME	eNodeB ↔ MME	Signalling
S1-U	eNodeB ↔ S-GW	Data transfer
S5/S8	S-GW ↔ P-GW	Data + control
S6a	MME ↔ HSS	Authentication
SGi	P-GW ↔ Internet	External connectivity

◆ 4. Traffic Types

- **User Traffic** → Actual data (internet, calls)
 - **Signalling Traffic** → Control messages
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◆ 5. Special Concepts (from diagram)

- **Local Breakout** → Data goes directly to internet (faster)
 - **Home Routed Traffic** → Data routed through home network (secure)
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Quick Summary (Exam Tip)

- **eNodeB** → Radio communication
 - **MME** → Control & authentication
 - **S-GW** → Data routing
 - **P-GW** → Internet connection
 - **HSS** → User database
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